

The Discriminant of a Cubic

A natural extension of the quadratic discriminant

Roots of polynomial equations: symmetric functions and the nature of the roots

Colour key: *green* = verification check | *blue* = aside / wider context.

1. The quadratic, rewritten in terms of its roots

For $ax^2 + bx + c = 0$ with roots α, β , the familiar discriminant is $\Delta = b^2 - 4ac$. The whole story — real, repeated, or complex roots — is told by its *sign*. The reason becomes transparent once we write Δ in terms of the roots themselves. Using $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$,

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = \frac{b^2}{a^2} - \frac{4c}{a} = \frac{b^2 - 4ac}{a^2}, \quad \text{so} \quad \boxed{\Delta = a^2(\alpha - \beta)^2}.$$

Since $a^2 > 0$, the sign of Δ is exactly the sign of $(\alpha - \beta)^2$:

$\Delta > 0 \Leftrightarrow$ roots real and distinct, $\Delta = 0 \Leftrightarrow$ equal roots, $\Delta < 0 \Leftrightarrow$ complex conjugate pair.

The last case is the key idea to carry forward: if $\alpha, \beta = u \pm iv$ then $(\alpha - \beta)^2 = (2iv)^2 = -4v^2 < 0$. A conjugate pair forces a negative squared difference.

2. The general principle

The expression $(\alpha - \beta)^2$ has two properties that make it the “right” object:

- it is **symmetric** in the roots, so it can be written using the coefficients alone (no need to solve the equation);
- it **vanishes precisely when two roots coincide**.

For a polynomial of any degree we keep both properties by multiplying together the squared differences of *every* pair of roots. For a cubic $ax^3 + bx^2 + cx + d = 0$ with roots α, β, γ this gives

$$\boxed{\Delta = a^4(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2}.$$

The power a^4 (rather than a^2) is what is needed to clear all denominators and leave a polynomial in a, b, c, d ; in general a degree- n polynomial carries the factor a^{2n-2} .

3. Why the sign classifies the roots

A real cubic has either three real roots or one real root with a complex conjugate pair. Take the product of squared differences in each case (the factor $a^4 > 0$ never affects the sign).

Three distinct real roots. Every difference is a non-zero real number, so every squared difference is positive and $\Delta > 0$.

One real root r and a conjugate pair $z, \bar{z} = u \pm iv$ ($v \neq 0$). Group the three factors:

$$(z - \bar{z})^2 = (2iv)^2 = -4v^2 < 0, \quad (r - z)^2(\bar{z} - r)^2 = [(r - z)(\bar{z} - r)]^2 = ((r - u)^2 + v^2)^2 > 0,$$

because $r - z$ and $\bar{z} - r = -\overline{(r - z)}$ are conjugate up to sign, so their product is the real number $-|r - z|^2$, which squares to $|r - z|^4$. One negative factor times one positive factor gives $\Delta < 0$.

A repeated root. Two roots coincide, a squared difference is 0, so $\Delta = 0$.

sign of Δ	nature of the roots of the real cubic
$\Delta > 0$	three distinct real roots
$\Delta = 0$	a repeated root (two or three roots equal)
$\Delta < 0$	one real root and a complex conjugate pair

This is the exact analogue of the quadratic rule — the same conjugate-pair mechanism, now sitting inside a product of three squared differences.

4. The discriminant in terms of the coefficients

Expanding the symmetric product and writing it through the coefficients gives, for ax^3+bx^2+cx+d ,

$$\Delta = 18abcd - 4b^3d + b^2c^2 - 4ac^3 - 27a^2d^2.$$

It is almost always cleaner to remove the x^2 term first. Substituting $x = t - \frac{b}{3a}$ reduces any cubic to the *depressed* form $t^3 + pt + q$, whose discriminant is the compact

$$\Delta = -4p^3 - 27q^2.$$

The sign rule of Section 3 applies unchanged, since the substitution only shifts the roots and so leaves every difference $\alpha - \beta$ — and hence Δ — untouched.

Aside — where this sits in the course. The cubic discriminant is not itself an examinable formula in 9231; it is included here to show that “ $b^2 - 4ac$ ” is one instance of a single idea — a symmetric, root-difference function whose sign decides how many roots are real. The same construction continues for quartics and beyond.

5. Worked examples

(a) Three distinct real roots. $x^3 - 6x^2 + 11x - 6 = 0$, i.e. $a = 1, b = -6, c = 11, d = -6$:

$$\Delta = 18(1)(-6)(11)(-6) - 4(-6)^3(-6) + (-6)^2(11)^2 - 4(1)(11)^3 - 27(1)^2(-6)^2 = 4 > 0,$$

so three distinct real roots — indeed $(x-1)(x-2)(x-3) = 0$.

(b) One real root and a conjugate pair. $x^3 + x + 1 = 0$ is depressed with $p = 1, q = 1$:

$$\Delta = -4(1)^3 - 27(1)^2 = -31 < 0,$$

so exactly one real root and a complex conjugate pair.

(c) A repeated root. $x^3 - 3x + 2 = 0$ is depressed with $p = -3, q = 2$:

$$\Delta = -4(-3)^3 - 27(2)^2 = 108 - 108 = 0,$$

so a repeated root — indeed $(x-1)^2(x+2) = 0$.

Check. (a) Roots 1, 2, 3 give $(\alpha - \beta)^2(\beta - \gamma)^2(\gamma - \alpha)^2 = (1)(1)(4) = 4 = \Delta$ ✓.
 (b) The real root is $r \approx -0.6823$; the cubic has no other real solution, matching $\Delta < 0$ ✓.
 (c) The factorisation $(x-1)^2(x+2)$ has a repeated root, matching $\Delta = 0$ ✓.